2020 AMC 10A Problem 4 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 4. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 4

Problem:

A driver travels for 2 hours at 60 miles per hour, during which her car gets 30 miles per gallon of gasoline. She is paid $\$0.50$ per mile, and her only expense is gasoline at $\$2.00$ per gallon. What is her net rate of pay, in dollars per hour, after this expense?

Answer Choices:

A. 20
B. 22
C. 24
D. 25
E. 26

Solution:

Since the driver travels 60 miles per hour and each hour she uses 2 gallons of gasoline, she spends $\$4$ per hour on gas. If she gets $\$0.50$ per mile, then she gets $\$30$ per hour of driving. Subtracting the gas cost, her net rate of money earned per hour is $\boxed{(E)\ 26}$.

$$\textbf{OR}$$

The driver is driving for 2 hours at 60 miles per hour, she drives 120 miles. Therefore, she uses

$$\frac{120}{30} = 4 \text{ gallons of gasoline}$$

So, total she has

$$\$0.50 \cdot 120 - \$2.00 \cdot 4 = \$60 - \$8 = \$52$$

So, her rate is

$$\frac{52}{2} = \boxed{(E)\ 26}.$$